Exercises: Logic

1. Propositional and First-Order Logic

1. Determine whether the following statements are **tautologies**, **contradictions**, or **contingent**:
   a) $P \vee \neg P$

   b) $(P \vee Q) \wedge (\neg P \vee \neg Q)$

   c) $(P \to Q) \vee (Q \to P)$

   d) $(P \wedge Q) \to P$
2. Construct the **truth table** for the following expressions:
   a) $(P \vee Q) \to R$

   b) $(P \wedge Q) \vee (\neg P \wedge \neg Q)$

   c) $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$
3. Translate the following statements into **propositional logic**:
   a) "If it is raining, then the ground is wet."

   b) "Either Alice or Bob will win the competition, but not both."

   c) "If the alarm is not working, then either there is a power failure or the battery is dead."

---

2. Formalization of Arguments

4. Determine whether the following arguments are **valid** using **truth tables** or **logical inference rules**:

   a) **Premises:**
      - $P \to Q$
      - $Q \to R$
      - P

      Conclusion: R

   b) **Premises:**
      - $P \vee Q$
      - $\neg P$

      Conclusion: Q

   c) **Premises:**

- $(P \to Q) \land (Q \to R)$
- $\neg R$
  Conclusion: $\neg P$

5. Use **natural deduction** to prove the following statements:
   a) $P \lor (Q \land R) \vdash (P \lor Q) \land (P \lor R)$
   b) $(P \to Q) \land (R \to S), P \lor R \vdash Q \lor S$

---

3. Applications in Mathematics and Computation

6. **Set Theory Application:** Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{2, 4, 6, 8\}$, and $B = \{1, 2, 3, 4\}$. Compute:
   a) $A \cup B$
   b) $A \cap B$
   c) A^c (the complement of A in U)
   d) $(A \cup B)^c$

7. **Boolean Algebra and Circuits:** Simplify the following Boolean expressions using **Boolean algebra laws**:
   a) $(A \lor B) \land (\neg A \lor B)$
   b) $(A \land B) \lor (\neg A \land B)$
   c) $A \lor (A \land B)$

8. **Logic Programming (Prolog):** Given the facts:

```
parent(alice, bob).
parent(bob, charlie).
parent(charlie, david).
```

   a) Write a **recursive rule** for the "ancestor" relation.
   b) Query whether **Alice is an ancestor of David** using your rule.